Build a filterable 20-university matrix comparing term structures, break lengths, and ECTS vs US credit ratios

Computer
Build a US-to-ECTS credit conversion calculator with degree-planning models and syllabus requirement breakdowns
Computer
Add a converter for UK CATS and Australian credits
Test the calculator logic with edge-case inputs
Export the matrix data to a different file format

Built an updated interactive HTML planner with the 20-university matrix, US/ECTS/UK CATS conversion, configurable Australian-credit conversion, degree-planning scenarios, and a syllabus workload breakdown. The matrix is also attached in **Excel** and JSON formats; all 30 calculator/model tests passed.

Matrix and exports

The HTML matrix supports:

- University and keyword search.
- Region and term-structure filters.
- Filtering by whether a numeric break length is available.
- Fall/autumn, spring/winter, summer, local credit loads, and conversion-ratio columns.
- Exporting filtered results as JSON.

The Excel workbook includes all 20 universities, column filters, frozen headings, and a “Read me” sheet documenting limitations.

Break data is deliberately bounded: eight universities have a numeric break example; the other 12 retain unknown lengths. Each row contains one selected break—not a complete vacation schedule. Date-based counts include both endpoints, and the dataset identifies the applicable year or scope.

Credit ratios remain blank where an institution-specific equivalency has not been verified. The matrix does not treat MIT units, quarter units, or every university’s local credits as US semester hours.

Conversion rules

Conversion	Implemented rule
US semester hours → ECTS	Multiply by a configurable ratio; default 2
ECTS → US semester hours	Divide by that ratio
UK CATS → ECTS	Divide by 2
ECTS → UK CATS	Multiply by 2
Australian local credits ↔ ECTS	Require a named institution/policy and an explicitly entered factor

The US default is a planning convention, not an automatic transfer award. The receiving institution determines accepted credit. [1]

UK guidance commonly uses 2 CATS per ECTS, with one UK credit representing approximately 10 notional learning hours. ECTS normally represents 25–30 workload hours per credit, so the credit conversion is not an exact equality of workload hours. [2] [3]

The Australian converter has **no** universal default. Melbourne's normal 50-point half-year and Sydney's normal 24-point semester illustrate why local point scales cannot share an unquestioned numerical ratio. Those normal loads alone do not authorize a transfer conversion. [4] [5]

Degree and syllabus models

The degree planner accepts a target, officially completed credits, pending transfer credits, and a planned term load—all in the same selected credit system.

It returns two separate scenarios:

- Official baseline: remaining credits and nominal terms using completed credits only.
- Conditional scenario: remaining credits and nominal terms if pending transfer is approved.

For example, the illustrative inputs of 120 required, 60 completed, 15 pending, and 15 per term produce four nominal terms at baseline—or three if the pending credit is approved. This is arithmetic planning, not a graduation determination: prerequisites, course availability, residency, grades, and program requirements still need validation.

The syllabus section provides manual fields for learning outcomes, prerequisites, materials, assessments, and four workload categories:

- Contact hours.
- Independent study.
- Assignments/projects.
- Exams/preparation.

It checks the total against the 25–30-hours-per-ECTS range. It does not extract a real syllabus, establish course equivalency, or verify academic requirements. [3]

Test results

Test group	Cases	Result
Conversion directions and custom Australian factors	8	Passed
Fractions, zero, invalid inputs, overflow, round trip	13	Passed
Degree-planning scenarios and validation	4	Passed
Syllabus workload boundaries and validation	5	Passed
Total	30	Passed

Tests ran against the actual JavaScript calculation functions. The embedded HTML script also passed syntax validation.

The attached JSON test report records the results. Browser interaction, accessibility, spreadsheet rendering, and download-button end-to-end behavior were not tested; this is a local prototype, not a deployed Semester capability.

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1. https://registrar.vanderbilt.edu/international/international-credit.php
2. https://www.qaa.ac.uk/docs/qaa/quality-code/higher-education-credit-framework-for-england.pdf
3. https://education.ec.europa.eu/sites/default/files/document-library-docs/ects-users-guide_en.pdf
4. https://students.unimelb.edu.au/course-admin/planning-your-course-and-subjects/study-load
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6. https://creditequivalenceguide.pages.ontraport.net/
7. https://clickcollege.com/online-courses-and-qualifications/cats-and-ects-credits-explained-what-do-40-120-240-and-360-credits-mean
8. https://www.scholaro.com/db/News/How-much-are-ECTS-CATS-and-SAQA-credits-worth-240
9. https://openeducat.org/articles/international-student-credit-transfer-guide/
10. https://blog.course-description.com/slug-what-is-the-ects-credit-system-and-how-does-it-work-3/
11. https://www.educations.com/student-resources/ects-to-us-credits-conversion
12. https://www.tcd.ie/study/assets/PDF/ECTS_Equivalents_Credit_Table_UK.pdf
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16. https://policy.unimelb.edu.au/MPF1293/
17. https://www.imperial.ac.uk/staff/tools-and-reference/quality-assurance-enhancement/programme-information/national-frameworks-and-credit-systems/
18. https://en.wikipedia.org/wiki/Credit_Accumulation_and_Transfer_Scheme

19. https://euler.euclid.int/policy-basis-of-ects-credits-and-us-euclid-conversions/